PG No. 42 $C_{3\mathrm{i}}(c)$ $\bar{3}$ [trigonal]

Table 1: Representation matrices

Irrep.						
A_g	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}1\end{bmatrix}$
$E_g^{(a)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$
$E_g^{(b)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
A_u	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-1\end{bmatrix}$
$E_u^{(a)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
$E_u^{(b)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$